

EXAM 7 – SPRING 2014

8. (4.5 points)

Given the following information as of December 31, 2013:

- An insurer had written only homeowners insurance until it began writing workers compensation on January 1, 2012.
- Five independent models were run for workers compensation and one independent model was run for homeowners, producing the following modeled coefficients of variation for accident year ultimate loss:

	<u>Model 1</u>	<u>Model 2</u>	<u>Model 3</u>	<u>Model 4</u>	<u>Model 5</u>
Homeowners	5%	-	-	-	-
Workers Compensation	21%	9%	45%	N/A	38%

- Model 4 failed to produce valid predictors for workers compensation.
- The claims adjusting process for homeowners has recently become automated.
- The company uses a third party to adjust medical claims for workers compensation. Claim information is received from the third party on a quarterly basis.
- In-house homeowners claims staff meet regularly with the valuation actuary.

a. (1.5 points)

Identify and define the three main components of internal systemic risk when determining liability risk margins for a book of business.

b. (1.5 points)

Identify one potential risk indicator for each line of business under each risk component listed in part a. above.

c. (1.5 points)

Briefly discuss whether each indicator in part b. above would receive a more or less favorable score on a balanced scorecard.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 8	
TOTAL POINT VALUE: 4.5	LEARNING OBJECTIVE: A5: Describe the various sources of risk and uncertainty that are associated with the determination of reserves. Calculate risk margins that consider these sources of risk and uncertainty.
SAMPLE ANSWERS	
Part a: 1.5 points	
<i>Sample Answer 1</i> Specification error – reflects the fact that it is not possible to precisely model the insurance process Parameter selection error – reflects the fact that it is not possible to perfectly estimate linear predictors or, in general, that model parameters are incorrect Data error – risk that there is an inadequate amount of data to build a model and/or that there is a lack of knowledge regarding pricing and underwriting practices <i>Sample Answer 2</i> Specification error – this is the risk of error in our inability of perfectly modelling the insurance process Parameter selection error – this is the uncertainty of selecting the best predictors or trends to describe the insurance process Data error – this is the risk that comes with inaccurate data or the lack of knowledge of our portfolio	
Homeowners Specification (b and c): .25 (b) .25 (c) point	
<i>Sample Answer 1</i> The number of model and simulations used to model the policy liabilities: not favorable since only uses one independent model instead of several Number of models run: less favorable – prefer to run more than one for homeowners to capture event/cat risk as well as provide check on first model validity Only one model was run, so could indicate difficulty specifying models: less favorable. Only 1 model gives less comfort model specifies correctly, since cannot compare results	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

Are our predictors stable over time: not favorable because claims adjusting process recently changed thus creates some uncertainty for the future

Meet with actuary: favorable since meeting regularly to share insights, beneficial to both parties

Meeting w/ claims staff regularly, could get expert opinion on parameters: more favorable – meeting with claims staff regularly helps get expert opinion, better estimates of parameters, lower risk of parameter error

Sample Answer 3

Is the data timely, reliable and consistent?: favorable since valuation actuary seems to meet regularly with claims staff and so seems to have a good knowledge and understanding of the portfolio.

Claim system automation: middle – probably favorable going forward for standard data process, but may be unfavorable as all parties adjust to new system

The claims process is automated, could lower data errors: more favorable – automated claims systems give less chance of data input errors

Workers Compensation Specification (b and c): .25 (b) .25 (c) point

Sample Answer 1

Since WC is a long tail coverage, do we have confidence in our testing of the model, has it been tested thoroughly to make sure it is appropriate? Stable model over model?: not favorable since the results of the independent model are very variable. Thus need to account for this with a higher risk margin.

Number of models run: favorable – multiple models provide balance and checks on one another

4 Models were run: more favorable – w/ 4 working models because they can compare results to test reasonability

Sample Answer 2

Have we selected the best predictors and trends to use in our model: not favorable since this is a new line of business and predictors are likely not yet perfect. Need some experience before this can be favorable.

SAMPLE ANSWERS AND EXAMINER'S REPORT

Amount of data (history) + level of experience in line of business: less favorable – new line for this company, may be some lacking knowledge thus far, good to use 4 models to balance this

High variance of CoV and one model failed to produce valid predictors. Could indicate parameter selection error: less favorable – the high CoV could indicate high parameter variance, showing more risk to parameter selection error

Sample Answer 3

Since insurer receives data quarterly, is the information received timely and reliable?: not favorable since this is a new line of business and the insurer does not have a very good knowledge of portfolio yet. Also receives claims information on a quarterly basis which may not be timely. Should also verify model 4 to see if error was not due to underlying data.

Claim information source: less favorable – good that data is received quarterly, but third party introduces risks with handoffs and quality

Third party claims staff, less direct control of data/tests: less favorable – third party claims staff gives less control over data/data checks, could result in errors

EXAMINER'S REPORT

Candidates were expected to be able to list the 3 main components of internal systemic risk. A vast majority of candidates were able to do this successfully.

Parts b and c required candidates to tie the facts given in the problem to each of the main components of internal systemic risk. While none of the line/component combinations individually were difficult to provide, it was challenging for many candidates to find specific and relevant examples for all 6 combinations. Candidates were expected to make some of these connections. Familiarity with the potential risk indicators enabled those candidates with higher scores in providing clear and comprehensive responses.

A significant number of candidates attempted to support their given risk indicator with facts not stated in the problem. This is considered a mistake under part c; however, in this situation the applicable part b solution would be acceptable if the candidate stated a general risk indicator listed in the paper even if that general risk indicator was unable to be supported with the given facts.

Part a

SAMPLE ANSWERS AND EXAMINER'S REPORT

With respect to identifying the risk types, Model or Specification error was acceptable however "Process error" was not an acceptable answer.

Also, for the Data error (Data accuracy) risk type, defining data error as solely the risk of having little data or low credibility was not acceptable; however, candidates were not penalized for the inclusion of this statement as a part of a broader explanation.

Parts b and c (Homeowner Specification)

There wasn't a common mistake in these parts. Most candidates correctly identified the number of models tested as an appropriate risk indicator.

Candidates who used the meetings with the claims staff as part of their response needed to state what the benefit of the meetings has from a parameter error perspective. Simply stating "meetings with claims staff" under part b with a more favorable score under part c was not an acceptable response

Another common mistake in this section was using the "long history of the line" as a risk indicator. This is a weak argument for a favorable score on its own and furthermore wasn't a stated fact in the problem.

Similarly, candidates who used the claims adjusting process as a risk indicator needed to support their response. See the rubric for examples.

Parts b and c (Workers Compensation Specification)

There wasn't a common mistake in these parts and most candidates correctly identified one of the possible risk indicators.

Interactions with the third party administrator (TPA) were not stated in the problem, so using the expertise or knowledge sharing of the TPA as support for a risk indicator would be a mistake under part c. As stated above however, acceptable answers under part b would include anything that discusses "best indicators", stable parameters, the value of the parameters selected, etc.

A common mistake was using the volume of workers compensation data as a risk indicator. Merely stating that the data was not credible due to the limited time the company had written the WC line was not sufficient for these parts. Instead candidates needed to discuss the potential for data coding errors and untimely data receipts from the TPA and/or a lack of complete knowledge of the portfolio and consequent misunderstanding of data as potential risk indicators.